

Homework 6: The Bootstrap and Missing Data

BEE 4850/5850

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 Note

All of the code used in this homework is accessible in the hw06.ipynb file in my GitHub repository.

Problem 1: Salamanders & the Parametric Bootstrap

1.1

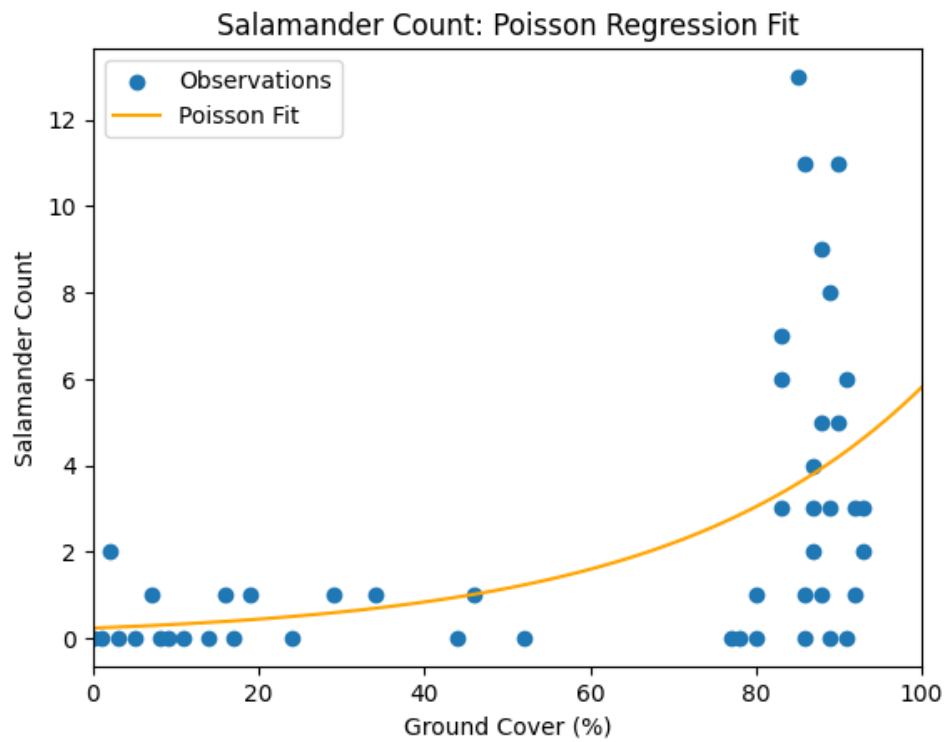


Figure 1: Scatterplot of Salamander Count vs. Ground Cover, with Poisson Regression fit shown in orange.

The intercept from the Poisson Regression fit is -1.478, and the coefficient from the fit is 0.032.

1.2

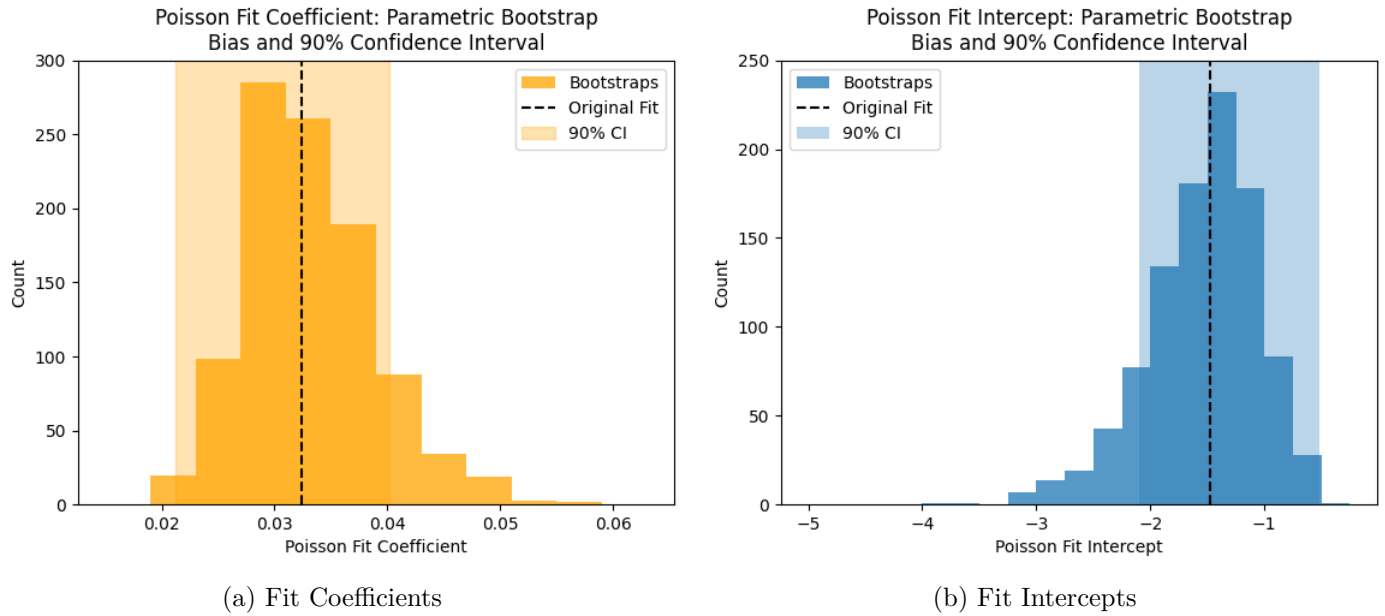


Figure 2: Histogram of Parametric Bootstrapped Poisson Fit a. coefficients and b. intercepts, with the coefficient from the fit in Figure 1 shown dashed in black. 90% Confidence interval highlighted in the background.

For the coefficient, the bias is -0.0008, and the 90% Basic Bootstrap CI is (0.039, 0.215). For the intercept, the bias is 0.0721, and the 90% Basic Bootstrap CI is (-0.522, -2.075).

1.3

Based on the above plots, there is relatively low uncertainty about the influence of ground cover on the expected number of salamanders. The 90% CI ranges are not very wide for either the intercept or the coefficient. More importantly, the CI is entirely above 0, which means that the influence of a 1% increase in ground cover on the number of salamanders is always positive. We can also verify that the data is likely a representative sample, since both statistics fall well within the bootstrap CI, and the bias is low for both.

Problem 2: Norfolk Tide Gauge & the Structured Bootstrap

2.1

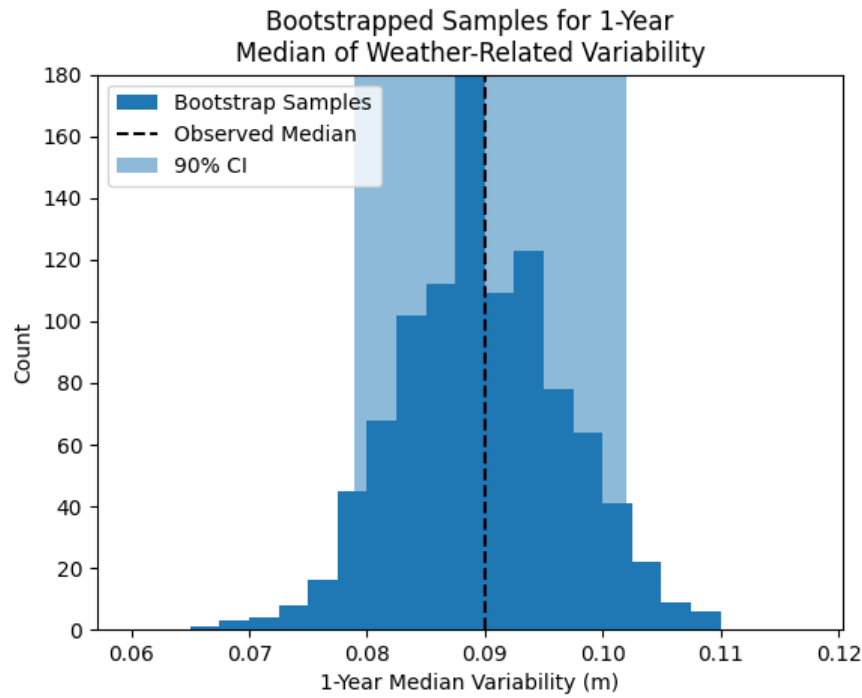


Figure 3: Histogram of length-20 rolling block bootstrap median weather-related tide variability, with the observed median shown in black. The 90% Confidence Interval is highlighted in the background.

The bias in the estimator is -0.00027, and the 90% CI is (0.078, 0.103).

2.2

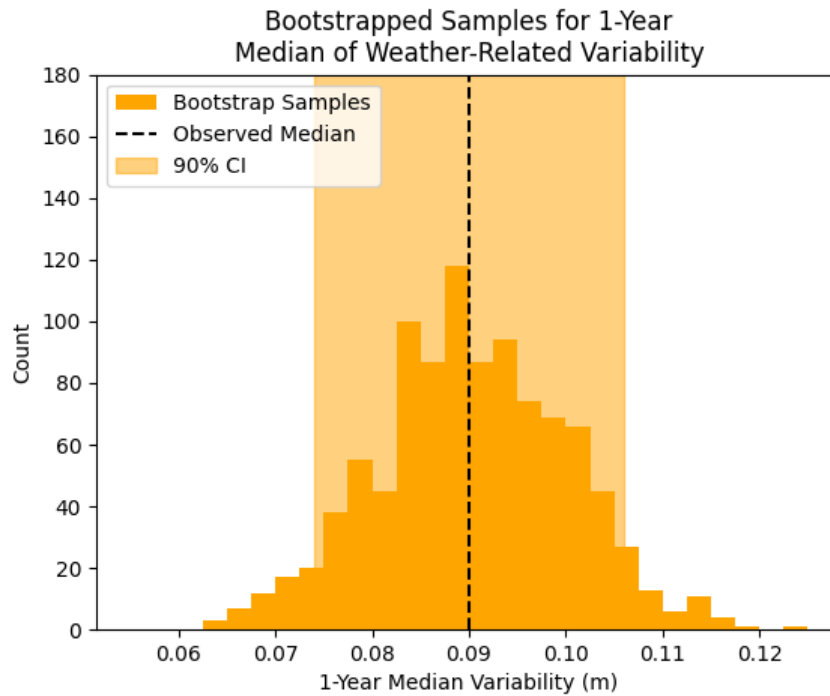


Figure 4: Same as in Figure 3, but with length-50 block lengths.

The bias in the estimator is 0.00083, and the 90% CI is (0.073 0.106). The bias is still very low/near-zero, although it did increase slightly. The confidence interval widened, with the low-end decreasing by 0.003 and the high-end increasing by 0.005.

2.3

Since the block sizes are larger, there are fewer blocks in each bootstrapped time series, which makes the bootstrap sample medians more susceptible to drawing a block with an outlier median. These extreme blocks would have a larger effect on the bootstrap sample medians in a length-50 bootstrap case than a length-20 case, thus explaining why the CI expanded.

Problem 3: Chicago Deaths & Missing Data

3.1

The `death`, `o3median`, `time`, and `tmpd` columns have no missing data. For the others, there are:

- 251 missing entries in `pm10median`
- 4387 missing entries in `pm25median`
- 27 missing entries in `so2median`

3.2

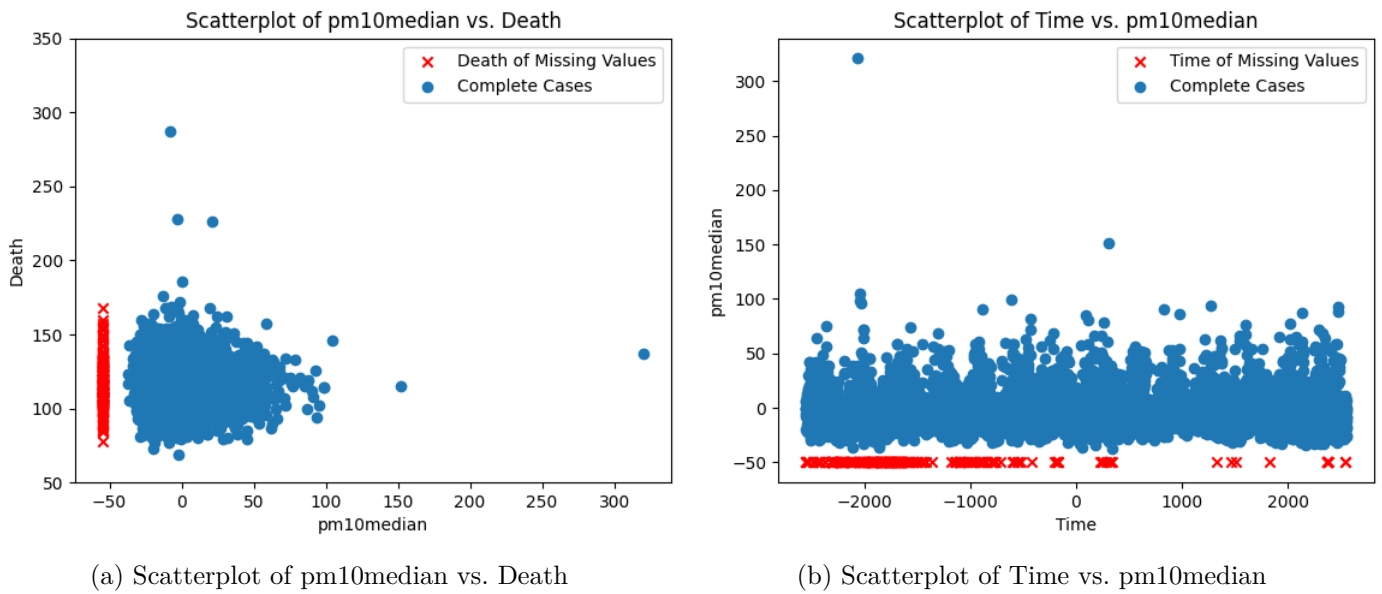


Figure 5: Exploratory Data Analysis of missing values in the Chicago deaths dataset.

In Figure 5a, the missing values for `pm10median` appear to cover the distribution of `death` (with the exception of the very high outliers), which suggests that the missingness is not dependent on the y variable (`death` in this case, since we have been focusing on predicting `death` with the environmental data). However, in Figure 5b, the missing values seem to come disproportionately from times earlier in the dataset. Since time is left unmodeled when attempting to predict `death` from `pm10median`, yet influences both, proceeding with complete-case analysis will likely lead to skewed results due to this time dependence of missing values. Thus, the `pm10median` data is Missing Not At Random.